Computer Vision

Unit 1 — Complete Study Notes

Topics Covered

• Image Formation
• Applications of Computer Vision (Drug Discovery, Medical & Defense, Criminal Detection)
• Homography & Projection
• Fourier Transform
• Convolution & Filtering
• Chessboard Distance
• Feature Descriptors
• Histogram Processing
• and supporting topics: Enhancement, Restoration, Compression, Morphology, Segmentation, Geometric & Intensity Transformations

1. Image Formation

1.1 Pixel and Pixel Grid

A digital image is composed of a finite grid of small elements called **pixels** (picture elements), arranged in rows and columns. Each pixel holds one or more numeric values that represent the brightness/colour of that tiny point of the scene. The collection of all pixel locations forms the **pixel grid** — the 2D array on which the image is sampled.

1.2 Image Representation

An image is mathematically represented as a 2-D function f(x, y), where x and y are spatial (row, column) coordinates and the amplitude of f at any point (x, y) gives the intensity (grey level) or colour of the image at that point. When x, y and the amplitude values are all finite, discrete quantities, the image is called a **digital image**.

1.3 How is a Digital Image Formed?

Image formation is the process by which a continuous real-world scene (continuous in both space and light intensity) is converted into a discrete digital image that a computer can store and process. It broadly involves:

- **Illumination** — light source(s) illuminate the scene.
- **Reflection/Transmission** — objects in the scene reflect or transmit light.
- **Sensing (Image Acquisition)** — a lens focuses light onto a sensor (CCD/CMOS) that converts the optical signal into an electrical signal.
- **Sampling** — the continuous signal is measured at discrete spatial locations (the pixel grid).
- **Quantization** — the continuous intensity value at each sample is mapped to one of a finite set of discrete grey/colour levels (e.g. 0–255 for 8-bit).

> ***Note:*** *Digital image acquisition therefore consists of two fundamental operations:* ***Sampling*** *(spatial digitization) and* ***Quantization*** *(amplitude digitization).*

1.4 Grey-scale and Colour Images

Grey-scale Image	Colour Image
Each pixel stores only ONE intensity value.	Uses three primary colours: Red (R), Green (G), Blue (B).
0 → Black, 255 → White (8-bit).	Each pixel stores three numbers, P(R,G,B).
Represents brightness only.	P(255,0,0) = Pure Red; P(255,255,255) = White.

1.5 Resolution of a Digital Image

Resolution = number of pixels present in an image. Example: 640 × 480 = 307,200 pixels. Higher resolution means more pixels, better detail, but a larger file size.

2. Applications of Computer Vision

Computer Vision (CV) techniques — image enhancement, restoration, segmentation, feature extraction and pattern recognition — are widely applied across scientific, industrial, medical, defense and security domains. Three important application areas covered in this unit are described below.

2.1 Drug Discovery

Computer vision plays a growing role in pharmaceutical research and drug discovery:

- **High-content screening** — automated microscopy combined with CV algorithms analyses thousands of cell images to study how candidate drug molecules affect cell shape, growth, and behaviour.
- **Molecular / crystal structure imaging** — CV helps analyse protein crystal images and molecular docking visualisations to identify promising drug compounds.
- **Pathology image analysis** — automated analysis of tissue and histopathology slide images speeds up detection of disease markers relevant to drug targets.
- **Toxicity and phenotypic profiling** — CV extracts quantitative features (shape, texture, intensity) from cell images to flag toxic or ineffective compounds early, reducing cost and time in the drug pipeline.

2.2 Medical and Defense Sector Applications

Medical Sector:

- Enhancement and restoration of X-ray, CT and MRI images for accurate diagnosis.
- Tumour/lesion detection and segmentation from scan images.
- Image registration to align scans taken at different times (e.g., cancer progression tracking).
- Computer-aided diagnosis (CAD) systems that flag abnormalities automatically.
- Surgical navigation and robotic-assisted surgery using real-time visual feedback.

Defense Sector:

- Surveillance and reconnaissance — object/vehicle/personnel detection from satellite, drone and CCTV footage, including enhancement of low-light or foggy imagery.
- Target recognition and tracking in autonomous weapons and guided systems.
- Camouflage and threat detection using pattern recognition.
- Autonomous navigation for unmanned vehicles (UAVs, UGVs) using scene understanding, obstacle detection and geometric transformations.
- Satellite image analysis for terrain mapping and strategic monitoring.

2.3 Criminal Detection — Applications

Computer vision is central to modern forensic and law-enforcement systems. Two of the most common biometric identification approaches are:

(a) Criminal Identification by Face

- **Face detection** locates a face within an image/video frame (e.g., using Haar cascades, HOG, or deep CNN-based detectors).
- **Feature extraction** captures distinguishing facial landmarks/geometry (eyes, nose, jaw line, distances between landmarks) as a feature descriptor / embedding.
- **Matching** compares the extracted features against a criminal-record database (1:N identification) or against a specific suspect (1:1 verification) using similarity/distance measures.
- Used in CCTV surveillance, airport security, and real-time suspect tracking.

(b) Criminal Identification by Thumb/Fingerprint

• Fingerprint images are pre-processed (enhancement, binarization, thinning) to make ridge patterns clear.

• **Minutiae extraction** — key points such as ridge endings and bifurcations are detected as feature descriptors.

• The extracted minutiae pattern is matched against the Automated Fingerprint Identification System (AFIS) database.

• Fingerprint-based identification is highly reliable because ridge patterns are unique to each individual and remain unchanged through life.

Note: *Both face-based and thumb/fingerprint-based criminal identification rely on the same general Computer Vision pipeline: Image Acquisition → Preprocessing/Enhancement → Feature Extraction (Feature Descriptors) → Matching/Classification against a stored database.*

3. Image Processing — Core Operations

Digital Image Processing (DIP) is the process of performing operations on a digital image using a computer to improve its quality, extract useful information, or prepare it for further analysis.

> **If f(x,y) = input image, then g(x,y) = T[f(x,y)] where T = image processing operation, g(x,y) = processed output image.**

T may represent enhancement, restoration, segmentation, filtering, compression, or feature extraction.

3.1 Image Enhancement

Improves the visual quality of an image (brightening, sharpening, contrast enhancement, equalization, intensity adjustment). Applications: medical imaging (X-ray/CT/MRI clarity), satellite imaging (roads/vegetation visibility), surveillance (low-light CCTV), autonomous vehicles (fog/rain/low-light visibility).

Techniques:

- Spatial Domain — Point Processing (brightness, contrast stretching, negative, log & power-law/gamma transformation, thresholding); Histogram-based (Equalization, Adaptive Histogram Equalization/AHE, CLAHE, Histogram Matching); Spatial Filtering (smoothening, sharpening).
- Frequency Domain — Low-Pass Filter (LPF), High-Pass Filter (HPF), Band-Pass Filter (BPF), Wavelet filtering.

3.2 Image Restoration

Recovers the original image from a degraded image by removing/reducing distortions (noise, blur, motion blur, sensor imperfections) using mathematical models. Causes of degradation: camera shake (motion blur), out-of-focus camera (defocus blur), sensor noise, dust/scratches, low-light conditions.

Problem	Restoration Method
Noise	Denoising
Motion Blur	Deblurring
Defocus Blur	Deconvolution
Missing Pixels	Image Inpainting

- Noise removal: Median Filter, Wiener Filter, Non-Local Means, Deep Learning Denoisers.
- Deblurring: Inverse Filtering, Wiener Filtering, Blind Deconvolution.
- Inpainting: fills damaged/missing regions using surrounding information.
- Super-resolution: reconstructs a high-resolution image from one or more low-resolution images.

3.3 Colour Image Processing & Colour Models

Model	Description
RGB	Red, Green, Blue — digital cameras, monitors, computer vision.
CMYK	Cyan, Magenta, Yellow, Black — used in colour printing.
HSV	Hue, Saturation, Value — close to human perception; used in segmentation.
YCbCr	Separates brightness (Y) from colour (Cb, Cr); used in image/video compression.
Lab (CIELAB)	Designed to match human visual perception; used in colour analysis.

3.4 Image Compression

Reduces image file size by removing redundant/unnecessary information while maintaining acceptable quality. Needed because large files require more storage, bandwidth, and processing time.

Types of redundancy removed:

- Spatial Redundancy — similar values among neighbouring pixels.
- Coding Redundancy — more efficient encoding of pixel data.
- Psycho-visual Redundancy — removal of details imperceptible to the human eye.

Lossless Compression	Lossy Compression
No information lost; perfectly reconstructible.	Higher compression ratio; slight, often imperceptible quality loss.
Examples: PNG, TIFF, GIF.	Example: JPEG.
Used in: medical imaging, scientific imaging, document storage.	Used in: digital photography, websites, social media, multimedia.

3.5 Morphological Operations

Techniques that modify the shape, size and structure of objects in an image using a small predefined pattern called a **structuring element (kernel)**. Unlike filtering (which modifies pixel intensities), morphological operations focus on the geometric structure of objects.

- Erosion
- Dilation
- Opening
- Closing

3.6 Segmentation

Partitions an image into meaningful regions corresponding to different objects or structures. Methods include:

- Thresholding
- Edge-based segmentation
- Region growing
- Watershed transformation
- Deep learning-based segmentation

4. Basic Relationships Between Pixels

Each pixel has a spatial relationship with its neighbouring pixels. These relationships let computers analyse image structure, detect object boundaries, identify connected regions, and extract meaningful information. Most algorithms analyse neighbouring pixels, local intensity variations and spatial connectivity — not pixels in isolation.

4.1 Adjacency, Path and Connectivity

- **Adjacency** — two pixels are adjacent if they are neighbours (4-adjacency, 8-adjacency, or m-adjacency) and satisfy a similarity criterion (e.g., same intensity range).
- **Path** — a sequence of pixels connecting one pixel to another where consecutive pixels are adjacent.
- **Connectivity** — pixels p and q in a subset S are connected in S if a path exists between them consisting entirely of pixels in S. For a pixel p in S, the set of all pixels connected to it forms a **connected component**. If S has only one connected component, S is a **connected set**.

4.2 Distance Measures

A distance measure calculates the distance between two points in an image. For pixels p, q, s with coordinates (x, y), (u, v), (w, z), D is a valid distance function/metric if it satisfies the standard metric axioms (non-negativity, symmetry, triangle inequality). Distance measures are used in segmentation, morphological operations, edge detection, object tracking and pattern recognition.

4.3 Types of Distance

Distance	Formula (between p(x,y) and q(u,v))	Description
Euclidean	De = sqrt[(x-u)^2 + (y-v)^2]	Straight-line (shortest) distance between two points.
City Block (Manhattan)	D4 = |x-u| + |y-v|	Sum of absolute horizontal & vertical distances (4-connectivity).
Chessboard	D8 = max(|x-u| , |y-v|)	Maximum of horizontal & vertical distance (8-connectivity); like a

4.4 Worked Example

Let A = (1,1) and B = (5,4).

- Euclidean Distance = sqrt[(5-1)^2 + (4-1)^2] = sqrt(16+9) = sqrt(25) = 5
- City Block Distance = |5-1| + |4-1| = 4 + 3 = 7
- Chessboard Distance = max(|5-1| , |4-1|) = max(4,3) = 4

> ***Note:*** *The **Chessboard distance (D8)** gets its name because it equals the minimum number of king moves needed to go from one square to another on a chessboard — diagonal moves count the same as horizontal/vertical moves.*

5. Image Interpolation & Basic Mathematical Operations

5.1 Image Interpolation

Interpolation estimates unknown pixel values using known values of nearby pixels. It is used when an image is enlarged (zoomed), reduced (shrunk), rotated, or geometrically transformed. Example: an original 500×500 image enlarged to 750×750 needs new pixel values computed from the existing ones via interpolation.

5.2 Elementwise (Pixel-by-Pixel) Operations

Each pixel is processed independently of others: Addition, Subtraction, Multiplication, Division.

Advantages	Disadvantages
Simple and fast; easy to implement; good for brightness/contrast changes.	Ignores neighboring pixels; cannot remove noise effectively; cannot detect ed

5.3 Linear vs Non-linear Operations

- **Linear** — output changes proportionally with input (Addition, Weighted Average, Linear Filtering).
- **Non-linear** — output does not change proportionally with input (Maximum, Minimum, Threshold, Median).

6. Geometric Transformations

Geometric transformations modify the spatial arrangement of pixels in an image — position, size, orientation, or shape.

6.1 Single-Pixel Transformation

Modifies digital image intensities individually using a transformation function s = T(z), where z is the original intensity and s is the mapped intensity (e.g., generating the image negative).

6.2 Translation

Shifting every pixel by a fixed distance along the horizontal/vertical direction, without changing orientation, size or shape.

> **P(x, y) → P'(x + tx, y + ty)**

Example: A pixel at P(4,5) translated by tx=3, ty=2 → P'(7,7). Real-life examples: moving an image on a slide, shifting a satellite image for alignment, repositioning an object in editing software, aligning video frames.

6.3 Scaling

Changes the size of an object by multiplying coordinates by scaling factors Sx (x-axis) and Sy (y-axis). Scale factor > 1 → larger; < 1 → smaller.

Scaling about an arbitrary point (e.g., the object's centre) is done in 3 steps: (1) translate the centre to the origin, (2) apply scaling, (3) translate back. Real-life examples: zooming a photo, Google Maps zoom, enlarging MRI/CT images.

6.4 Rotation

Rotates every point of an object through a specified angle about a fixed centre of rotation (usually the origin, unless specified otherwise). Real-life examples: smartphone portrait↔landscape rotation, Google Maps navigation rotation, rotating CT/MRI scans, robot camera image correction, de-skewing scanned documents before OCR.

6.5 Reflection

Every point of an object is mapped to the opposite side of a specified axis of reflection, maintaining the same perpendicular distance from that line (the line acts like a mirror). Real-life examples: selfie camera mirrored preview, horizontal/vertical flipping in Photoshop, data augmentation for ML, medical image visualisation.

6.6 Shearing (Skewing)

One coordinate of every point is displaced proportionally to the other coordinate, causing the object to slant while preserving parallelism of lines.

6.7 Comparison of Transformations

Property	Translation	Scaling	Rotation	Reflection	Shearing
Changes Position	Yes	No	Yes	Yes	Yes
Changes Size	No	Yes	No	No	No
Changes Orientation	No	Non-uniform may distort	Yes	Yes (reversed)	Yes (slanted)

Preserves Distance	Yes	No	Yes	Yes	No
Preserves Angles	Yes	Uniform only	Yes	Yes	No
Preserves Shape	Yes	Uniform only	Yes	Yes	No
Preserves Parallel Lines	Yes	Yes	Yes	Yes	Yes

6.8 Affine Transformation

Combines multiple transformations (translation, rotation, scaling, shearing) into a single operation. It preserves straight lines and parallel lines, making it one of the most widely used geometric transformations. Applications: image registration, object tracking, image warping.

> **[x', y', 1]^T = A · [x, y, 1]^T — where A is the affine transformation matrix**

6.9 Degrees of Freedom (DOF)

DOF = number of independent parameters needed to uniquely define a transformation. More DOF → more flexible but more complex.

Transformation	Parameters	DOF
Translation	tx, ty	2
Rotation	θ	1
Scaling	Sx, Sy	2
Reflection	fixed axis (predefined)	0 (discrete choice)
Shearing	Shx, Shy	2

6.10 Homography

A **homography** is a projective transformation that maps points from one plane to another (e.g., mapping a planar surface as seen from one camera view to how it appears from a different viewpoint). It is represented by a 3×3 matrix H (with 8 independent degrees of freedom, since it is defined up to scale) that operates on points expressed in **homogeneous coordinates**:

> **[x', y', w']^T = H · [x, y, 1]^T, where the final image coordinates are (x'/w', y'/w')**

Unlike affine transformations, a homography can also model **perspective distortion** — it does not necessarily preserve parallel lines, but it does preserve straight lines (collinearity) and cross-ratios.

- Homography generalises translation, rotation, scaling, shearing and affine transforms — all of these are special cases of a homography.
- Estimated from at least 4 pairs of corresponding points between two images/planes.
- **Applications**: image stitching/panorama creation, perspective correction (e.g., correcting a photographed document to a top-down view), augmented reality (aligning virtual objects with a real planar surface), camera calibration, and homography-based image registration.

6.11 Orthogonal Transformation

A transformation represented by an orthogonal matrix (its transpose equals its inverse). Orthogonal transformations preserve lengths and angles — rotations and reflections are classic examples. They are important in image transform theory because orthogonal (or unitary) transform kernels preserve energy, which is a desirable property when converting an image into a different representation (e.g., a

frequency-domain basis).

6.12 Projection

Projection is the geometric process of mapping points from a higher-dimensional space (typically the 3-D world) onto a lower-dimensional space (the 2-D image plane) — this is fundamentally how a camera forms an image of a 3-D scene.

- **Perspective Projection** — models real camera imaging: 3-D world points are projected through a single centre of projection (the camera's optical centre) onto the image plane. Objects farther from the camera appear smaller (depth-dependent scaling), and parallel lines can appear to converge at a vanishing point — matching how human eyes and real cameras perceive the world.
- **Orthographic (Parallel) Projection** — projection rays are parallel to each other (no centre of projection). Object size in the image does not change with depth/distance. Commonly used in engineering drawings, CAD, and as a simplified camera model when the scene depth variation is small compared to the viewing distance.
- Projection is closely tied to **homography**: a homography describes the mapping produced when a planar scene is projected (perspectively) onto two different image planes (e.g., two camera views of the same flat surface).

7. Image Registration

Image registration aligns two or more images of the same scene. Given an **input image** and a **reference image**, the objective is to geometrically transform the input image to produce an output image that is aligned (registered) with the reference image. Example application: medical scans for cancer tracking, where scans taken at different times must be aligned to compare tumour growth precisely.

8. Intensity Transformations & Histogram Processing

8.1 Intensity (Gray-Level) Transformation

A spatial-domain operation that modifies the intensity value of every pixel according to a predefined mathematical function s = T(r), where r is the input intensity and s is the output intensity. Two broad types: **Linear** and **Non-linear**.

8.2 Linear Intensity Transformations

Output intensity changes proportionally to input intensity following a straight-line equation. The amount of change stays constant across the intensity range. Examples: brightness adjustment, contrast stretching, linear scaling. Used for uniform image modification.

8.3 Non-linear Transformations

The relationship between input and output intensity is not linear; the transformation curve bends up or down depending on the required enhancement. Useful for emphasising either dark or bright regions.

- **Image Negative**: s = L - 1 - r (for an L-level image; e.g., s = 255 - r for 8-bit). Reverses dark and light regions.
- **Log Transformation**: s = c·log(1+r). Expands dark pixel values while compressing the higher (bright) range — useful for enhancing detail in dark regions such as Fourier-spectrum display.
- **Inverse Log Transformation**: the reverse operation of log transform.
- **Power-Law (Gamma) Transformation**: s = c·r^γ. Provides flexible control over brightness/contrast via the gamma (γ) parameter.

8.4 Power-Law / Gamma Transformation — Effect Table

Gamma Value (γ)	Effect on Image
0.3	Very bright
0.5	Bright
0.8	Slightly bright
1	No change
1.5	Slightly dark
2	Dark
3	Very dark

Why Gamma Transformation is required: Camera sensors respond nonlinearly to light, display devices (monitors/TVs/projectors) have nonlinear brightness characteristics, and human eyes perceive brightness nonlinearly. Gamma correction compensates for these, improving contrast, revealing hidden

details, and adapting appearance for different display devices.

8.5 Thresholding

Converts a grayscale image into a binary image by comparing every pixel with a threshold value T. If intensity ≥ T → assigned one value (usually white/255, Foreground/Object); otherwise → the other value (usually black/0, Background).

Example: 8-bit image (0–255), T = 120. Pixels ≥ 120 become 255 (white); pixels < 120 become 0 (black).

Types of Thresholding

- **Global** — a single threshold value is applied to the whole image.
- **Adaptive** — the image is divided into regions, each with its own threshold (e.g., left side T=90, right side T=150); useful when lighting varies across the image.
- **Otsu's Method** — automatic threshold selection: analyses the image histogram and picks the threshold that best separates foreground and background by maximising between-class separation (equivalently, minimising within-class variance).

8.6 Contrast Stretching

Contrast is the difference in intensity between the darkest and brightest regions of an image. High contrast → large difference (sharp, easy to interpret); low contrast → small difference (dull, faded, washed-out).

General intensity-range transformation example: Map the input range 50–180 to the output range 20–220 (linear stretch of the useful intensity band to the full display range).

Types of Contrast Stretching

- **Linear Contrast Stretching** — a single straight-line mapping over the whole range.
- **Piecewise-Linear Contrast Stretching** — used when only a specific intensity region needs enhancement; uses multiple line segments defined by two control points, (r1,s1) and (r2,s2).

8.7 Histogram Processing (Histogram-Based Enhancement)

An image's **histogram** is a plot of the frequency (count) of each grey-level value occurring in the image — it reveals the overall distribution of brightness. Histogram processing techniques reshape this distribution to enhance the image:

- **Histogram Equalization** — redistributes pixel intensities so the output histogram is as flat/uniform as possible, spreading out the most frequent intensity values and thereby increasing global contrast, especially useful for images with a narrow intensity range (too dark or too bright overall).
- **Adaptive Histogram Equalization (AHE)** — applies histogram equalization locally over small regions (tiles) of the image rather than globally, improving local contrast where lighting is uneven.
- **CLAHE (Contrast Limited Adaptive Histogram Equalization)** — an improved AHE that clips the histogram at a predefined limit before equalization to avoid over-amplifying noise in near-uniform regions.
- **Histogram Matching (Specification)** — transforms an image's histogram to match a specified target histogram (e.g., to match the tonal look of a reference image).

8.8 Bit-Plane Slicing

Separates an image into different **bit planes** to analyse how much information each bit of every pixel contributes. Unlike intensity transformations (which modify gray-level values directly), bit-plane slicing

decomposes the binary representation of each pixel into individual layers — e.g., Bit Plane 7 (most significant, contributes most visual structure) down to Bit Plane 0 (least significant, mostly noise-like detail).

9. Image Transforms (incl. Fourier Transform)

An **Image Transform** is a mathematical operation that converts an image from the spatial domain into another domain, known as the **transform domain** or **frequency domain**. It simplifies image analysis, enhancement, compression, restoration, or feature extraction.

9.1 Why Transforms?

- Fast computation for certain operations.
- Conceptual insight into image content (e.g., which frequencies dominate).
- Obtaining transformed data as a compact measurement/representation.
- Efficient storage and transmission.

9.2 Key Steps in Using a Transform

- (1) Transform the image into the new domain.
- (2) Carry out the required task(s) in the transformed domain (e.g., filtering, compression).
- (3) Apply the inverse transform to return to the spatial domain.

9.3 Kernel Properties

- A kernel is **separable** if a 2-D transform kernel can be expressed as a product of two 1-D kernels (allowing the 2-D transform to be computed as two successive 1-D transforms — rows then columns — which is far more efficient).
- A kernel is **symmetric** if the separable 1-D kernels used for rows and columns are identical.

9.4 Types of Image Transforms

- Fourier Transform — Continuous FT, Discrete FT (DFT), Fast FT (FFT)
- Discrete Cosine Transform (DCT)
- Wavelet Transform
- Hadamard Transform
- Walsh Transform
- Haar Transform
- etc.

9.5 Fourier Transform

The Fourier Transform converts a spatial-domain signal/image into its frequency-domain representation, expressing the image as a sum (or integral) of sinusoidal basis functions of different frequencies, amplitudes, and phases. It reveals the frequency content of an image — smooth regions correspond to low frequencies and edges/fine detail correspond to high frequencies.

- **Continuous Fourier Transform (FT)** — converts a continuous spatial-domain function into its continuous frequency-domain representation. Mainly used for theoretical analysis and general signal processing since digital computers work with discrete data.
- **Discrete Fourier Transform (DFT)** and **Fast Fourier Transform (FFT)** — since digital images are discrete, DFT/FFT are the most commonly used forms in Digital Image Processing. FFT is a computationally efficient algorithm for calculating the DFT.

1-D Continuous FT: $F(u) = \int f(x) \cdot e^{-j2\pi ux}\, dx$ (integral over all x)

2-D Continuous FT: $F(u,v) = \iint f(x,y) \cdot e^{-j2\pi(ux+vy)}\, dx\, dy$

Applications of the Fourier Transform in image processing: frequency-domain filtering (removing noise or specific frequency patterns), image compression (identifying and discarding perceptually unimportant high-frequency components), analysis of periodic patterns/textures, and image restoration (e.g., deblurring using frequency-domain deconvolution).

9.6 Discrete Cosine Transform (DCT) — Example: JPEG Compression

Rather than storing millions of individual pixel values, JPEG compression stores only the important frequency coefficients using the **Discrete Cosine Transform (DCT)**. Since the human eye is less sensitive to high-frequency detail, many high-frequency DCT coefficients can be discarded/quantized with minimal perceptible loss — this is how image transforms simplify and enable efficient image processing tasks like compression.

10. Convolution and Filtering

10.1 What is Convolution?

Convolution is a fundamental spatial-domain operation in which a small matrix called a **kernel** (or filter/mask) is systematically slid over every pixel of an image. At each position, the kernel values are multiplied with the corresponding overlapping pixel values, and the results are summed to produce the new pixel value in the output (filtered) image.

$g(x,y) = \Sigma\, \Sigma\, f(x-i, y-j) \cdot h(i,j)$ (sum over the kernel window; h = kernel/filter)

In practice, most image-processing libraries implement **correlation** (kernel applied without flipping) but call it convolution; true convolution mathematically flips the kernel before sliding — for symmetric kernels the two operations give identical results.

10.2 Why Filtering / Convolution is Used

- Noise reduction (smoothing/blurring filters).
- Edge detection and sharpening (highlighting intensity changes).
- Feature extraction (detecting corners, blobs, textures for later analysis).
- Basis for Convolutional Neural Networks (CNNs) in modern computer vision.

10.3 Common Convolution/Spatial Filters

Filter Type	Purpose	Examples
Smoothing (Low-pass) filter	Blurs the image, reduces noise	Mean/Averaging filter, Gaussian filter, Median filter
Sharpening (High-pass) filter	Enhances edges and fine detail	Laplacian filter, Unsharp masking
Edge-detection filter	Highlights boundaries between regions	Sobel, Prewitt, Canny, Laplacian

10.4 Key Properties of Convolution

- **Kernel size** — determines the neighbourhood considered (e.g., 3×3, 5×5); larger kernels consider more context but cost more computation.
- **Padding** — how border pixels (where the kernel extends beyond the image) are handled (zero-padding, replication, mirroring).
- **Stride** — the step size by which the kernel moves across the image.

• Convolution is central to both classical filtering (smoothing, sharpening, edge detection) and to feature-learning layers in modern deep-learning-based computer vision (CNNs).

10.5 Frequency-Domain View of Filtering

By the Convolution Theorem, convolution in the spatial domain is equivalent to point-wise multiplication in the frequency domain (and vice versa). This is why Fourier-domain filtering (LPF, HPF, BPF — see Section 9) achieves the same goals as spatial convolution but can be computationally faster for large kernels, since the image and kernel only need to be transformed once, multiplied, and transformed back.

11. Feature Descriptors in Image Formation/Processing

11.1 What is a Feature?

In computer vision, a **feature** is a distinctive, informative piece of an image — such as an edge, corner, blob, or texture pattern — that can be reliably detected and used to describe, match, or recognise objects across different images, viewpoints, or lighting conditions.

11.2 What is a Feature Descriptor?

A **feature descriptor** is a numerical representation (typically a vector) that encodes the local appearance around a detected feature/keypoint, so that the same real-world point can be recognised and matched even when the image undergoes changes in scale, rotation, illumination, or viewpoint. The overall pipeline is: **Feature Detection** (locate interesting keypoints) → **Feature Description** (encode the local region around each keypoint as a descriptor vector) → **Feature Matching** (compare descriptors between images using a distance metric).

11.3 Desirable Properties of a Good Descriptor

- **Distinctiveness** — different features should produce clearly different descriptors.
- **Invariance** — should remain stable under rotation, scale, illumination and viewpoint changes.
- **Robustness** — tolerant to noise and minor image distortions.
- **Efficiency** — compact and fast to compute/match, especially for real-time applications.

11.4 Common Feature Descriptors

Descriptor	Key Idea
SIFT (Scale-Invariant Feature Transform)	Detects scale- and rotation-invariant keypoints; describes local gradient orientation histog
SURF (Speeded-Up Robust Features)	A faster approximation of SIFT using integral images and box filters.
ORB (Oriented FAST and Rotated BRIEF)	Combines the FAST keypoint detector with a rotation-aware BRIEF binary descriptor; fas
HOG (Histogram of Oriented Gradients)	Describes an image region using the distribution of local gradient directions; widely used
LBP (Local Binary Pattern)	Encodes texture by thresholding each pixel's neighbours against the centre pixel value; c
Corner descriptors (Harris, Shi-Tomasi)	Detect corner keypoints based on local intensity gradient variation in multiple directions.

11.5 Applications of Feature Descriptors

- Image stitching / panorama creation (matching keypoints across overlapping images).
- Object recognition and tracking.
- Face recognition and biometric identification (e.g., criminal detection by face — see Section 2.3).
- Fingerprint minutiae matching (criminal detection by thumb — see Section 2.3).
- 3-D reconstruction and camera pose/homography estimation (matched feature correspondences are used to compute the homography matrix — see Section 6.10).

12. Practical / Application Task (from Unit)

Task: Develop a computer vision application that allows a user to improve the visual quality of an image and apply different geometrical transformations. The application should display the original and processed images side by side and allow comparison of different techniques. Mention all required parameters.

Part A — Spatial-Domain Enhancement

- Brightness adjustment
- Contrast adjustment
- Image negative
- Log transformation
- Power-law / gamma transformation
- Thresholding
- Contrast stretching

Part B — Geometrical Transformations

- Translation
- Scaling
- Rotation
- Horizontal reflection
- Vertical reflection
- Reflection about the origin
- Shearing along the x-axis
- Shearing along the y-axis
- Affine transformation

Note: *These notes are compiled from the uploaded 'Module 1.pptx' slide deck (Dr. Neha Rathore, SCAI). A few listed topics — Drug Discovery, Medical & Defense CV applications, Criminal Detection, Homography, Projection, and Feature Descriptors — were present only as slide titles with blank/placeholder content in the uploaded file, so those sections above have been written out in full using standard Computer Vision course material so that no topic from your list is skipped.*